

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.65276	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.20898	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1438.19396	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.200191	0.200191	183.02154	0.772718	12.23

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.65276.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.20898.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1438.19396.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.216219.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.200191.
- Held-out runtime ms: 183.02154.
- Robustness dispersion across held-out structure families: 0.010162.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.268391, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.253555, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.230653, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.140754, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.107602, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite

Metric	Baseline Nearest Neighbor	Baseline Clarke-Wright Savings	Baseline Regret Insertion LS	Evolved Solver
Feasibility Rate	1.0	1.0	1.0	1.0
Heldout Feasible Gap	0.2734	**0.0738 (best)**	0.4664	0.2002
Heldout Runtime (ms)	42.65	77.21	1438.19	183.02
Robustness Dispersion	0.00278	0.00779	0.0869	0.01016

- ****Feasibility:**** All solvers, including the evolved one, achieved perfect feasibility (1.0) on held-out instances.
- ****Objective Gap:**** The evolved solver's heldout gap (0.2002) is better than nearest neighbor (0.2734) and regret insertion local search (0.4664), but clearly worse than Clarke-Wright savings (0.0738).
- ****Runtime:**** The evolved solver is slower than both nearest neighbor (42.7 ms) and Clarke-Wright (77.2 ms), but significantly faster than regret insertion local search (1438 ms).
- ****Robustness:**** The evolved solver's robustness dispersion (0.0102) is higher than nearest neighbor and Clarke-Wright but much lower than regret insertion local search.

Conclusion

While the evolved solver shows improvement over some baselines (nearest neighbor and regret insertion local search) in objective gap and maintains feasibility, it does ****not**** clearly outperform the best fixed baseline (Clarke-Wright savings), which remains superior in solution quality and runtime. The evolved solver offers a moderate tradeoff between runtime and solution gap but does not dominate the fixed baselines across all metrics.